



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Main)

FULL SYLLABUS

07-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE (ALL STAR BATCH)]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	C	D	B	A	D	C	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	D	D	D	D	B	C	A	B	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	2500	3	20	3	12	12	120	4	3	113

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	A	A	C	C	A	C	C	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	C	C	C	B	C	B	D	C	D
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	15	14	108	19	3	19	1	525	2218	4

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B	C	B	C	A	B	D	C	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	A	C	B	A	A	C	B	B	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	2	5	1	6	1	14	15	6	28	16

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (A)

Truth table for the given logic gate :

A B Y

0 0 1

0 1 1

1 0 1

1 1 0

The truth table is similar to that of a NOR gate.

2. Ans (C)

From energy conservation,

[after bullet gets embedded till the
system comes momentarily at rest]

$$(M + m)gh = \frac{1}{2}(M + m)v_1^2$$

[v_1 is velocity after collision]

$$\therefore v_1 = \sqrt{2gh}$$

Applying momentum conservation, (just before and just after collision)

$$mv = (M + m)v_1$$

$$v = \left(\frac{M + m}{m} \right) v_1$$

$$= \frac{6}{10 \times 10^{-3}} \times \sqrt{2 \times 9.8 \times 9.8 \times 10^{-2}}$$

$$\approx 831.55 \text{ m/s}$$

3. **Ans (C)**

Since force on a point charge by magnetic field is always perpendicular to $\vec{v}$ [$\vec{F} = q\vec{v} \times \vec{B}$]

$\therefore$ Work by magnetic force on the point charge is zero.

4. **Ans (D)**

The nature of flow is determined by Reynolds Number.

$$R_e = \frac{\rho v D}{\eta}$$

$$\left[\begin{array}{l} \rho \rightarrow \text{density of fluid} ; \quad \eta \rightarrow \text{coefficient of} \\ v \rightarrow \text{velocity of flow} \quad \quad \quad \text{viscosity} \\ D \rightarrow \text{Diameter of pipe} \end{array} \right]$$

From NCERT

If $R_e < 1000 \rightarrow$ flow is steady

$1000 < R_e < 2000 \rightarrow$ flow becomes unsteady

$R_e > 2000 \rightarrow$ flow is turbulent

$$R_{e \text{ initial}} = 10^3 \times \frac{0.18 \times 10^{-3}}{\pi \times (0.5 \times 10^{-2})^2 \times 60} \times \frac{1 \times 10^{-2}}{10^{-3}} = 382.16$$

$$R_{e \text{ final}} = 10^3 \times \frac{0.48 \times 10^{-3}}{\pi \times (0.5 \times 10^{-2})^2 \times 60} \times \frac{1 \times 10^{-2}}{10^{-3}} = 1019.09$$

5. **Ans (B)**

Given :

$$\vec{v} = 0.5t^2 \hat{i} + 3t \hat{j} + 9\hat{k}$$

$$\vec{v}_{\text{at } t=2} = 2\hat{i} + 6\hat{j} + 9\hat{k}$$

$\therefore$ Angle made by direction of motion of mosquito will be,

$$\cos^{-1} \frac{2}{11} \text{ (from x-axis)} = \tan^{-1} \frac{\sqrt{117}}{2}$$

$$\cos^{-1} \frac{6}{11} \text{ (from y-axis)} = \tan^{-1} \frac{\sqrt{85}}{6}$$

$$\cos^{-1} \frac{9}{11} \text{ (from z-axis)} = \tan^{-1} \frac{\sqrt{40}}{9}$$

None of the option is matching.

Hence this question should be bonus.

6. **Ans (A)**

$$\frac{2K\lambda}{r} = \frac{\sigma}{\epsilon_0} \quad (x=3m)$$

$$\sigma = 0.424 \times 10^{-9} \frac{C}{m^2}$$

7. **Ans (D)**

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}}$$

$$\frac{\lambda_1}{\lambda_2} = \sqrt{\frac{m_2}{m_1}}$$

$$\frac{\lambda_e}{\lambda_p} = \sqrt{\frac{m_p}{m_e}} = \sqrt{1831.4} = 42.79$$

8. **Ans (C)**

$$V_s = \frac{P}{i} = \frac{60}{0.11} = 545.45$$

$$V_p = 220$$

$$V_s > V_p$$

$\Rightarrow$ Step up transformer

9. **Ans (B)**

$$A = \lambda N$$

$$N = nN_A \quad \left(t_{1/2} = \frac{\ln 2}{\lambda} \right)$$

$$N = \left(\frac{1.5 \times 10^{-3}}{198} \right) N_A$$

$$A = \left(\frac{\ln 2}{t_{1/2}} \right) N$$

$$1 \text{ Curie} = 3.7 \times 10^{10} \text{ Bq}$$

$$A = 365 \text{ Bq}$$

10. **Ans (D)**

$$\lambda = \frac{RT}{\sqrt{2\pi d^2 N_A P}}$$

$$\lambda = 102 \text{ nm}$$

11. **Ans (B)**

$$\frac{1}{F} = \left[\frac{\mu_L}{\mu_S} - 1 \right] \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\text{If } \mu_L = \mu_S \Rightarrow \frac{1}{F} = 0 \Rightarrow F = \infty$$

12. **Ans (D)**

$$Y = \frac{\text{Stress}}{\text{Strain}} = \frac{FL}{Al} = \frac{mg \cdot L}{\pi R^2 \cdot \ell}$$

$$\frac{\Delta Y}{Y} = \frac{\Delta m}{m} + \frac{\Delta L}{L} + 2 \cdot \frac{\Delta R}{R} + \frac{\Delta \ell}{\ell}$$

$$\frac{\Delta Y}{Y} \times 100$$

$$= 100 \left[\frac{1}{1000} + \frac{1}{1000} + 2 \left(\frac{0.001}{0.2} \right) + \frac{0.001}{0.5} \right]$$

$$= \frac{1}{10} + \frac{1}{10} + 1 + \frac{1}{5} = \frac{14}{10} = 1.4\%$$

13. Ans (D)

$$\alpha_A > \alpha_B$$

Length of both strips will decrease

$$\Delta L_A > \Delta L_B$$



14. Ans (D)

$$500 = (1.5)^2 \times R \times 20$$

$$E = (3)^2 \times R \times 20$$

$$E = 2000 \text{ J}$$

15. Ans (D)

Statement I :

$$v_{\max} = \sqrt{\mu R g} = \sqrt{(0.2) \times 2 \times 9.8}$$

$$v_{\max} = 1.97 \text{ m/s}$$

$$7 \text{ km/h} = 1.944 \text{ m/s}$$

Speed is lower than $v_{\max}$, hence it can take safe turn.

Statement II

$$v_{\max} = \sqrt{R g \left[\frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right]}$$

$$= \sqrt{2 \times 9.8 \left[\frac{1 + 0.2}{1 - 0.2} \right]} = 5.42 \text{ m/s}$$

$$18.5 \text{ km/h} = 5.14 \text{ m/s}$$

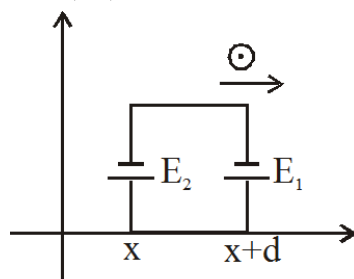
Speed is lower than $v_{\max}$, hence it can take safe turn.

16. Ans (B)

$$D = 2\sqrt{2Rh}$$

$$h = \frac{D^2}{8R} = \frac{45^2}{8 \times 6400} \text{ km} \approx 39.55 \text{ m}$$

17. Ans (C)



$$E_1 = \frac{B_0(x+d)}{a} v_0 d$$

$$E_2 = \frac{B_0(x)}{a} v_0 d$$

$$E_{\text{net}} = E_1 - E_2$$

$$E_{\text{net}} = \frac{B_0 v_0 d^2}{a}$$

18. Ans (A)

$$A = A_0 e^{-\gamma t} = A_0 e^{-\frac{bt}{2m}}$$

$$\frac{A_0}{2} = A_0 e^{-\frac{bt}{2m}}$$

$$\frac{bt}{2m} = \ln 2$$

$$t = \frac{2m}{b} \ln 2 = \frac{2 \times 500 \times 0.693}{20}$$

$$t = 34.65 \text{ second.}$$

19. Ans (B)

$$N_1 = N_0 e^{-\lambda t_1}$$

$$\frac{N_1}{N_0} = e^{-\lambda t_1}$$

$$0.67 = e^{-\lambda t_1}$$

$$\ln(0.67) = -\lambda t_1$$

$$N_2 = N_0 e^{-\lambda t_2}$$

$$\frac{N_2}{N_0} = e^{-\lambda t_2}$$

$$0.33 = e^{-\lambda t_2}$$

$$\ln(0.33) = -\lambda t_2$$

$$\ln(0.67) - \ln(0.33) = \lambda t_1 - \lambda t_2$$

$$\lambda(t_1 - t_2) = \ln\left(\frac{0.67}{0.33}\right)$$

$$\lambda(t_1 - t_2) \approx \ln 2$$

$$t_1 - t_2 \approx \frac{\ln 2}{\lambda} = t_{1/2}$$

$$\text{Half life} = t_{1/2} = 20 \text{ minutes.}$$

20. Ans (A)

Red light and blue light have different wavelength and different frequency.

PART-1 : PHYSICS

SECTION-II

1. Ans (2500)

$$Q = i^2 R T$$

$$R = \frac{Q}{i^2 t} = \frac{10 \times 10^{-3}}{4 \times 10^{-6} \times 1} = 2500 \Omega$$

2. Ans (3)

$$C = \frac{\epsilon_0 A}{\frac{d}{2K} + \frac{d}{2}} = \frac{2\epsilon_0 A}{\frac{d}{K} + d}$$

$$= \frac{2 \times 2\epsilon_0}{\frac{1}{3.2} + 1} = \frac{4 \times 3.2}{4.2} \epsilon_0$$

$$= 3.04 \epsilon_0$$

3. Ans (20)

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{r} = (2\hat{i}) - (2\hat{i} + 3\hat{j} + 4\hat{k}) = -3\hat{j} - 4\hat{k}$$

$$\& \vec{F} = 4\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -3 & -4 \\ 4 & 3 & 4 \end{vmatrix}$$

$$= \hat{i}(-12 + 12) - \hat{j}(0 + 16) + \hat{k}(0 + 12)$$

$$= -16\hat{j} + 12\hat{k}$$

$$\therefore |\vec{\tau}| = \sqrt{16^2 + 12^2} = 20$$

4. Ans (3)

$$\text{Energy given} = U_f - U_i$$

$$= 0 - \left(-\frac{3}{5} \frac{GM^2}{R} \right)$$

$$= \frac{3}{5} \frac{GM^2}{R}$$

$$x = 3$$

5. Ans (12)

$$\omega_1 = 0.02; \mu_1 = 1.5; \omega_2 = 0.03; \mu_2 = 1.6$$

Achromatic combination

$$\therefore \theta_{\text{net}} = 0$$

$$\theta_1 - \theta_2 = 0$$

$$\theta_1 = \theta_2$$

$$\omega_1 \delta_1 = \omega_2 \delta_2$$

$$\& \delta_{\text{net}} = \delta_1 - \delta_2 = 2^\circ$$

$$\delta_1 - \frac{\omega_1 \delta_1}{\omega_2} = 2^\circ$$

$$\delta_1 \left(1 - \frac{\omega_1}{\omega_2} \right) = 2^\circ$$

$$\delta_1 \left(1 - \frac{2}{3} \right) = 2^\circ$$

$$\delta_1 = 6^\circ$$

$$\delta_1 = (\mu_1 - 1) A_1$$

$$6^\circ = (1.5 - 1) A_1$$

$$A_1 = 12^\circ$$

6. Ans (12)

$$\vec{a} = \frac{\vec{F}}{m} = \frac{2\hat{i} + 3\hat{j} + 5\hat{k}}{2}$$

$$= \hat{i} + 1.5\hat{j} + 2.5\hat{k}$$

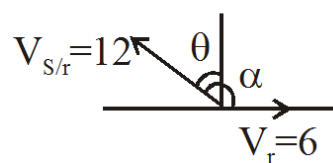
$$\vec{\tau} = \vec{u} t + \frac{1}{2} \vec{a} t^2$$

$$= 0 + \frac{1}{2} (\hat{i} + 1.5\hat{j} + 2.5\hat{k}) (16)$$

$$= 8\hat{i} + 12\hat{j} + 20\hat{k}$$

$$b = 12$$

7. Ans (120)



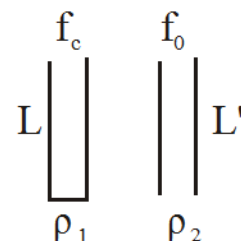
$$12 \sin \theta = v_r$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = 30^\circ$$

$$\therefore \alpha = 120^\circ$$

8. Ans (4)



$$f_c = f_0$$

$$\frac{3V_C}{4L} = \frac{2V_0}{2L'}$$

$$\frac{3V_C}{4L} = \frac{V_0}{L'}$$

$$L' = \frac{4L}{3} \frac{V_0}{V_C} = \frac{4L}{3} \sqrt{\frac{B \cdot \rho_1}{\rho_2 \cdot B}} \quad (B \text{ is bulk modulus})$$

$$= \frac{4L}{3} \sqrt{\frac{\rho_1}{\rho_2}}$$

$$x = 4$$

9. Ans (3)

$$a = \frac{g \sin \theta}{1 + \frac{1}{mR^2}} = \frac{g \sin \theta}{1 + \frac{1}{2}} = \frac{2}{3} g \sin \theta$$

$$b = 3$$

10. Ans (113)

$$n = 0.60 = 1 - \frac{T_L}{T_H}$$

$$\frac{T_L}{T_H} = 0.4 \Rightarrow T_L = 0.4 \times 400$$

$$T_L = 160 \text{ K} = -113^\circ\text{C}$$

$$|T| = 113$$

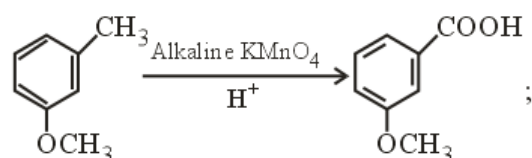
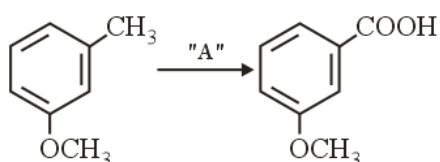
PART-2 : CHEMISTRY

SECTION-I

1. Ans (C)

The green house gases are CO_2 , $\text{H}_2\text{O}_{(\text{vapour})}$ & CH_4 .

2. Ans (C)



3. Ans (A)

Reduction of $\text{Al}_2\text{O}_3 \rightarrow \text{Al}$ is carried out by electrolytic reduction of its fused salts.

ZnO , Fe_2O_3 & Cu_2O can be reduced by carbon.

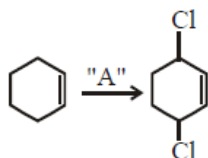
4. Ans (A)

$\text{Na} \rightarrow [\text{Ne}] 3s^1$ IE_1 is very low but IE_2 is very high due to stable noble gas configuration of Na^+ .

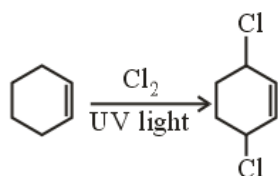
$\text{Mg} \rightarrow [\text{Ne}] 3s^2$ IE_1 & $\text{IE}_2 \rightarrow$ Low

IE_3 is very high.

5. Ans (C)



For substitution at allylic position in the given compound, the reagent used is $\text{Cl}_2/\text{uv light}$. The reaction is free radical halogenation.



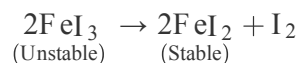
6. Ans (C)

The secondary structure of protein includes two type :

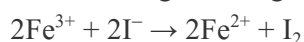
(a) α -Helix (b) β -pleated sheet

In α -Helix structure, the poly peptide chain is coil around due to presence of Intramolecular H-Bonding.

7. Ans (A)



Due to strong reducing nature of I^-



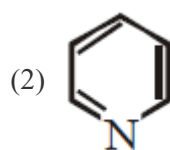
remaining halides of Fe^{2+} & Fe^{3+} are stable.

8. Ans (C)

Urea -HCHO resin is used in manufacture of wood laminates.

9. Ans (C)

(1) NaH (sodium Hydride) is used as a reducing reagent.



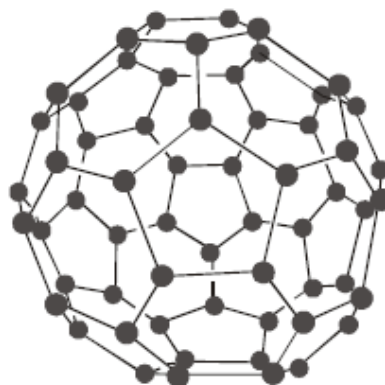
(2) In pyridine, due to free electron

on N atom, it is basic in nature.

Hence statement I is false & II is true.

10. Ans (D)

Structure of C_{60}

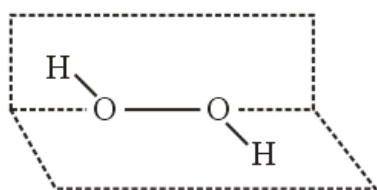


It contains 20 hexagons and 12 pentagons

12

so option 4 is incorrect.

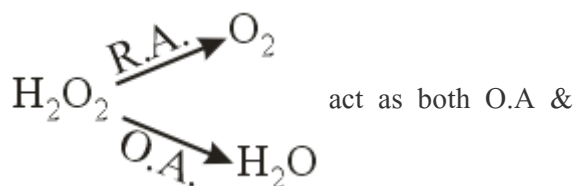
11. Ans (B)



Structure of H_2O_2

(Open book type) $\rightarrow$ Non planar

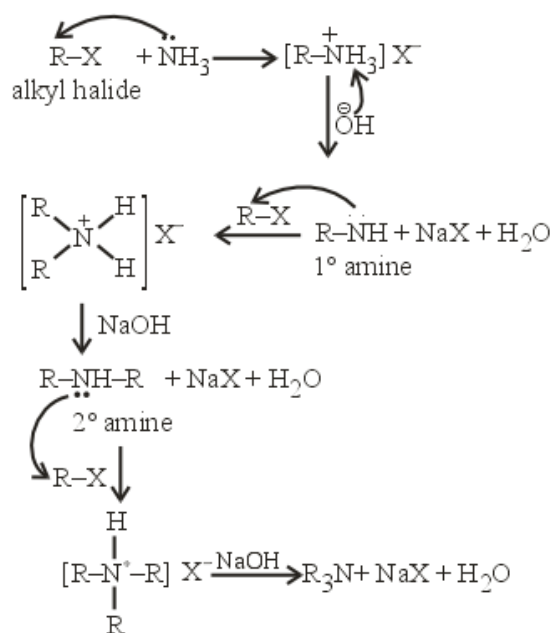
H_2O_2 is used in the treatment of effluents.



R.A.

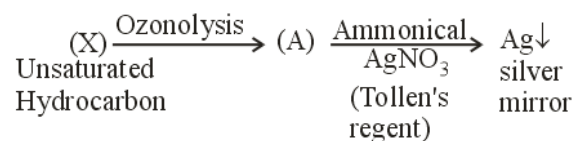
H_2O_2 is miscible in water due to hydrogen bonding.

12. Ans (C)



So the purpose of NaOH in the above reactions is to remove acidic impurities.

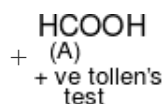
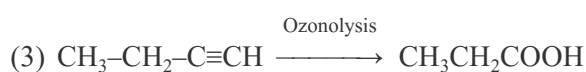
13. Ans (C)



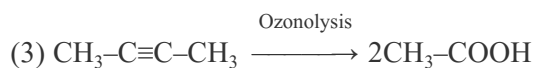
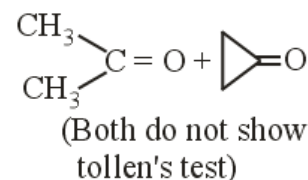
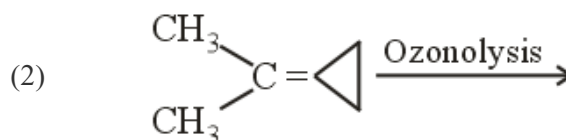
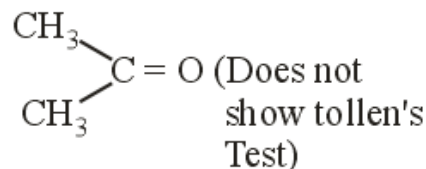
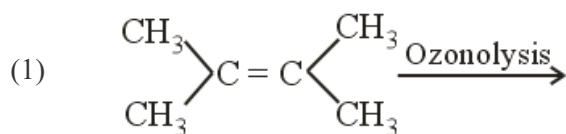
As (A) compound given positive tollen's test hence it may consist $-CHO$ (aldehyde group).

or it can be $HCOOH$

So for the given option :



and for other compounds (options):



(Does not show tollen's test)

14. Ans (C)

For the given compounds :

(1) $\text{CH}_3-\text{C}(=\text{O})-\text{NH}-\text{C}_2\text{H}_5$; L.P. on Nitrogen is delocalised.

(2) $\text{CH}_3\text{CH}_2-\text{N}(\text{CH}_2\text{CH}_3)_2$; ; L.P. on Nitrogen is localised.

(3) $\text{CH}_3-\text{C}(=\text{O})-\text{NH}-\text{C}(=\text{O})-\text{CH}_3$; L.P. on Nitrogen is delocalised due to conjugation with both $-\text{C}(=\text{O})-$
(Hence least basic)

(4) $\text{CH}_3-\text{CH}_2-\text{NH}-\text{CH}_2-\text{CH}_3$; L.P. on Nitrogen is localised.

15. Ans (B)

X = $_{33}\text{As}$ → Metalloid

Y = $_{53}\text{I}$ → Nonmetal

Z = $_{83}\text{Bi}$ → Metal

16. Ans (C)

Match list :-

(a) Lassaigne's Test	(iii)	N, S, P and Halogen
(b) Cu(II) Oxide	(i)	Carbon
(c) AgNO_3	(iv)	Halogen specifically.
(d) Sodium fusion extract given black precipitate with acetic acid and lead acetate ($\text{CH}_3\text{COOH}/(\text{CH}_3\text{COO})_2\text{Pb}$)	(ii)	Sulphur

Option-(a)-(iii) ; (b)-(i) ; (c)-(iv) ; (d)-(ii)

17. Ans (B)

- * Colloidal solution exhibits colligative properties
- * An ordinary filter can not stop the flow of colloidal particles.
- * Flocculating power increases with increase the opposite charge of electrolyte.
- * Colloidal particles show brownian motion.

18. Ans (D)

(a) $_{58}\text{Ce} \rightarrow [\text{Xe}]4f^2 5d^0 6s^2$

In complex $\text{Ce}^{4+} \rightarrow [\text{Xe}] 4f^0 5d^0 6s^0$
there is no unpaired electron so $\mu_m = 0$

(b) $_{64}\text{Gd}^{3+} \rightarrow [\text{Xe}]4f^7 5d^0 6s^0$

contain seven unpaired electrons so,

$$\mu_m = \sqrt{7(7+2)} = \sqrt{63} \text{ B. M.}$$

(c) $_{63}\text{Eu}^{3+} \rightarrow [\text{Xe}]4f^6 5d^0 6s^0$

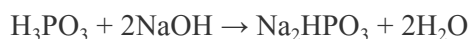
contain six unpaired electron

$$\text{so, } \mu_m = \sqrt{6(6+2)} = \sqrt{48} \text{ B. M.}$$

Hence, order of spin only magnetic movement

$$\boxed{b > c > a}$$

19. Ans (C)

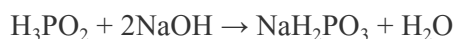


50 ml 1M

1M V = ?

$$\Rightarrow \frac{n_{\text{NaOH}}}{n_{\text{H}_3\text{PO}_3}} = \frac{2}{1}$$

$$\Rightarrow \frac{1 \times V}{50 \times 1} = \frac{2}{1} \Rightarrow \boxed{V_{\text{NaOH}} = 100 \text{ ml}}$$



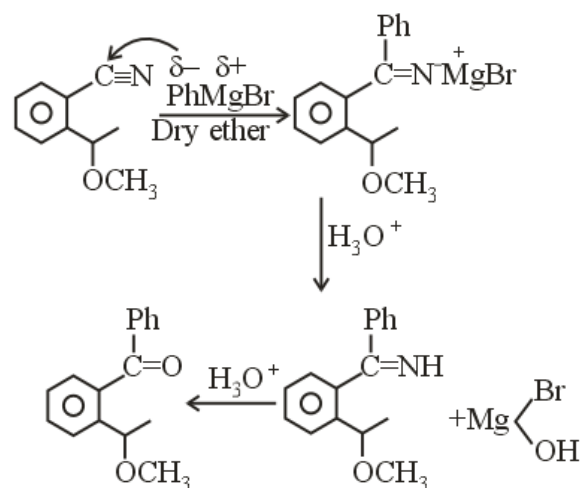
100 ml 1M

2M V = ?

$$\Rightarrow \frac{n_{\text{NaOH}}}{n_{\text{H}_3\text{PO}_3}} = \frac{1}{1}$$

$$\Rightarrow \frac{1 \times V}{2 \times 100} = \frac{1}{1} \Rightarrow \boxed{V_{\text{NaOH}} = 200 \text{ ml}}$$

20. Ans (D)



PART-2 : CHEMISTRY

SECTION-II

1. Ans (15)

HCP structure : Per atom, there will be one octahedral void (OV) and two tetrahedral voids (TV).

Therefore total three voids per atom are present in HCP structure.

→ therefore total no of atoms of Ga will be-

$$= \frac{\text{Mass}}{\text{Molar Mass}} \times N_A$$

$$= \frac{0.581\text{g}}{70\text{g/mol}} \times 6.023 \times 10^{23}$$

→ Now, total Number of voids = 3 × total no. of atoms

$$= 3 \times \frac{0.581}{70} \times 6.023 \times 10^{23}$$

$$= 14.99 \times 10^{21}$$

$$\approx 15 \times 10^{21}$$

2. Ans (14)

$$\text{Given conc}^n \text{ of KCl} = \frac{\text{m. mol}}{\text{L}}$$

$$\text{: Conductance (G)} = 0.55 \text{ mS}$$

$$\text{: Cell constant} \left(\frac{\ell}{A} \right) = 1.3 \text{ cm}^{-1}$$

To Calculate : Molar conductivity (λ_m) of sol.

$$\rightarrow \text{Since } \lambda_m = \frac{1}{1000} \times \frac{k}{m} \dots (1)$$

$$\rightarrow \text{Molarity} = 5 \times 10^{-3} \frac{\text{mol}}{\text{L}}$$

$$\rightarrow \text{Conductivity} = G \times \left(\frac{\ell}{A} \right) = 0.55 \text{ mS} \times \frac{1.3}{\frac{1}{100}} \text{ m}^{-1}$$

$$= 55 \times 1.3 \text{ mSm}^{-1}$$

$$\text{eq}^n (1) \lambda_m = \frac{1}{1000} \times \frac{55 \times 1.3}{\left(\frac{5}{1000} \right)} \frac{\text{mSm}^2}{\text{mol}}$$

$$\Rightarrow \lambda_m = 14.3 \frac{\text{mSm}^2}{\text{mol}}$$

3. Ans (108)

$$\text{Given } t_2 = 54 \text{ min} \quad T_{1/2} = 18 \text{ min}$$

A

B

$$t = 0 \text{ 'x' M} \quad t = 0 \text{ 'x' M}$$

$$\Rightarrow \text{To calculate : } [A_t] = 16 \times [B_t] \dots (1) \text{ time} = ?$$

$$\Rightarrow \text{For I order kinetic : } [A_t] = \frac{A_0}{(2)^n}$$

$n \rightarrow$ no of Half lives

$\Rightarrow$ Now from the relation (1)

$$[A_t] = 16 \times [B_t]$$

$$\Rightarrow \frac{x}{(2)^{n_1}} = \frac{x}{(2)^{n_2}} \times 16$$

$$\Rightarrow (2)^{n_2} = (2)^{n_1} \times (2)^4$$

$$\Rightarrow n_2 = n_1 + 4$$

$$\Rightarrow \frac{t}{(t_{1/2})_2} = \frac{t}{(t_{1/2})_1} + 4$$

$$\Rightarrow t \left(\frac{1}{18} - \frac{1}{54} \right) = 4 \Rightarrow t = \frac{4 \times 18 \times 54}{36}$$

$$\Rightarrow \boxed{t = 108 \text{ min}}$$

4. Ans (19)

In Duma's method of estimation of Nitrogen.

0.1840 gm of organic compound gave 30 mL of nitrogen which is collected at 287 K & 758 mm of Hg.

Given ;

Aqueous tension at 287 K = 14 mm of Hg.

Hence actual pressure = (758 - 14)

$$= 744 \text{ mm of Hg.}$$

$$\text{Volume of nitrogen at STP} = \frac{273 \times 744 \times 30}{287 \times 760}$$

$$V = 27.935 \text{ mL}$$

$$\therefore 22400 \text{ mL of } N_2 \text{ at STP weighs} = 28 \text{ gm.}$$

$$\therefore 27.94 \text{ mL of } N_2 \text{ at STP weighs}$$

$$= \left(\frac{28}{22400} \times 27.94 \right) \text{ gm}$$

$$= 0.0349 \text{ gm}$$

$$\text{Hence \% of Nitrogen} = \left(\frac{0.0349}{0.1840} \times 100 \right) = 18.97 \%$$

Rond off. Answer = 19 %

5. **Ans (3)**

For, $n = 5$

$$\ell = (0, 1, 2, 3, 4)$$

If $\ell = 0, m = 0$

$$\ell = 1, m = \{-1, 0, +1\}$$

$$\ell = 2, m = \{-2, -1, 0, +1, +2\}$$

$$\ell = 3, m = \{-3, -2, -1, 0, +1, +2, +3\}$$

$$\ell = 4, m = \{-4, -3, -2, -1, 0, +1, +2, +3, +4\}$$

5d, 5f and 5g subshell contain one-one orbital having $m_\ell = +2$

6. **Ans (19)**

$$\text{Given } P_A^0 = 21 \text{ kPa} \Rightarrow P_B^0 = 18 \text{ kPa}$$

→ An Ideal solution is prepared by mixing 1 mol A and 2 mol B.

$$\rightarrow X_A = \frac{1}{3} \text{ and } X_B = \frac{2}{3}$$

→ Acc to Raoult's law

$$P_T = X_A P_A^0 + X_B P_B^0$$

$$\Rightarrow P_T = \left(\frac{1}{3} \times 21\right) + \left(\frac{2}{3} \times 18\right)$$

$$\Rightarrow P_T = 7 + 12 = 19 \text{ KPa}$$

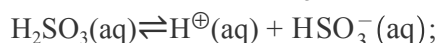
7. **Ans (1)**

H_2SO_3 [Dibasic acid]

$$c = 0.588 \text{ M}$$

⇒ pH of solution ⇒ due to First dissociation only since $K_{a1} \gg K_{a2}$

⇒ First dissociation of H_2SO_3



$$K_{a1} = 1.7 \times 10^{-2}$$

$t = 0 \text{ C}$

$t \quad \text{C-x} \quad \quad \quad x \quad \quad \quad x$

$$\Rightarrow K_{a1} = \frac{1.7}{100} = \frac{[\text{H}^+][\text{HSO}_3^-]}{[\text{H}_2\text{SO}_3]}$$

$$\Rightarrow \frac{1.7}{100} = \frac{x^2}{(0.58 - x)}$$

$$\Rightarrow 1.7 \times 0.588 - 1.7x = 100 x^2$$

$$\Rightarrow 100x^2 + 1.7x - 1 = 0$$

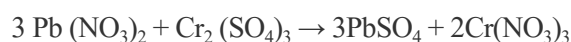
$$\Rightarrow [\text{H}^+] = x = \frac{-1.7 + \sqrt{(1.7)^2 + 4 \times 100 \times 1}}{2 \times 100}$$

$$= 0.09186$$

Therefore pH of sol. is : $\text{pH} = -\log [\text{H}^+]$

$$\Rightarrow \text{pH} = -\log (0.09186) = 1.036 \approx 1$$

8. **Ans (525)**



$$35 \text{ ml} \quad 20 \text{ ml}$$

$$0.15 \text{ M} \quad 0.12 \text{ M}$$

$$= 5.25 \text{ m.mol} = 2.4 \text{ m.mol} \quad 5.25 \text{ m.mol}$$

$$= 5.25 \times 10^{-3} \text{ mol}$$

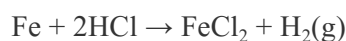
$$\text{therefore moles of PbSO}_4 \text{ formed} = 5.25 \times 10^{-3}$$

$$= 525 \times 10^{-5}$$

9. **Ans (2218)**

$$T = 298 \text{ K}, R = 8.314 \frac{\text{J}}{\text{mol K}}$$

→ Chemical reaction is



$$50 \text{ g} \quad P = 1 \text{ bar}$$

$$= \frac{50}{55.85} \text{ mol}$$

→ Work done for 1 mol gas

$$= -P_{\text{ext}} \times \Delta V$$

$$= \Delta n g RT$$

$$= -1 \times 8.314 \times 298 \text{ J}$$

$$\rightarrow \text{Work done for } \frac{50}{55.85} \text{ mol of gas}$$

$$= -1.8314 \times 298 \times \frac{50}{55.85} \text{ J}$$

$$= -2218.059 \text{ J}$$

$$\approx -2218 \text{ J}$$

10. **Ans (4)**

$$\lambda_{\text{absorbed}} = 498 \text{ nm (given)}$$

The octahedral splitting energy

$$\Delta_0 \text{ or } E = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{498 \times 10^{-9}}$$

$$= 0.0399 \times 10^{-17} \text{ J}$$

$$= 3.99 \times 10^{-19} \text{ J}$$

$$= 4.00 \times 10^{-19} \text{ J (round off)}$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$$C_1 + C_2 \rightarrow C_1$$

$$\begin{vmatrix} 2 & 1 + \cos^2 x & \cos 2x \\ 2 & \cos^2 x & \cos 2x \\ 1 & \cos^2 x & \sin 2x \end{vmatrix}$$

$$R_1 - R_2 \rightarrow R_1$$

$$\begin{vmatrix} 0 & 1 & 0 \\ 2 & \cos^2 x & \cos 2x \\ 1 & \cos^2 x & \sin 2x \end{vmatrix}$$

Open w.r.t. R_1

$$-(2 \sin 2x - \cos 2x)$$

$$\cos 2x - 2 \sin 2x = f(x)$$

$$f(x)|_{\max} = \sqrt{1+4} = \sqrt{5}$$

2. Ans (B)

Total cases :

$$6 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2$$

$$n(s) = 6 \cdot 6!$$

Favourable cases :

Number divisible by 3 $\equiv$

Sum of digits must be divisible by 3

Case-I

$$1, 2, 3, 4, 5, 6$$

$$\text{Number of ways} = 6!$$

Case-II

$$0, 1, 2, 4, 5, 6$$

$$\text{Number of ways} = 5 \cdot 5!$$

Case-III

$$0, 1, 2, 3, 4, 5$$

$$\text{Number of ways} = 5 \cdot 5!$$

$$n(\text{favourable}) = 6! + 2 \cdot 5 \cdot 5!$$

$$P = \frac{6! + 2 \cdot 5 \cdot 5!}{6 \cdot 6!} = \frac{4}{9}$$

3. Ans (C)

$$\lim_{x \rightarrow 0^+} f(x) = f(0) = \lim_{x \rightarrow 0^-} (x)$$

$$\lim_{x \rightarrow 0^+} \frac{\cos^{-1}(1-x^2) \cdot \sin^{-1}(1-x)}{x(1-x)(1+x)}$$

$$\lim_{x \rightarrow 0^+} \frac{\cos^{-1}(1-x^2)}{x \cdot 1 \cdot 1} \cdot \frac{\pi}{2}$$

$$\text{Let } 1-x^2 = \cos \theta$$

$$\frac{\pi}{2} \lim_{x \rightarrow 0^+} \frac{\theta}{\sqrt{1-\cos \theta}}$$

$$\frac{\pi}{2} \lim_{\theta \rightarrow 0^+} \frac{\theta}{\sqrt{2} \sin \frac{\theta}{2}} = \frac{\pi}{\sqrt{2}}$$

$$\text{Now, } \lim_{x \rightarrow 0^-} \frac{\cos^{-1}(1-(1+x)^2) \sin^{-1}(-x)}{(1+x)-(1+x)^3}$$

$$\lim_{x \rightarrow 0^-} \frac{\frac{\pi}{2} (-\sin^{-1} x)}{(1+x)(2+x)(-x)}$$

$$\lim_{x \rightarrow 0^-} \frac{\frac{\pi}{2} \cdot \frac{\sin^{-1} x}{x}}{1 \cdot 2} = \frac{\pi}{4}$$

$$\Rightarrow \text{RHL} \neq \text{LHL}$$

Function can't be continuous

$\Rightarrow$ No value of α exist

4. Ans (B)

Plane passing through $(42, 0, 0)$, $(0, 42, 0)$, $(0, 0, 42)$

From intercept form, equation of plane is $x + y + z = 42$

$$\Rightarrow (x-11) + (y-19) + (z-12) = 0$$

$$\text{let } a = x-11, b = y-19, c = z-12$$

$$a + b + c = 0$$

Now, given expression is

$$3 + \frac{a}{b^2 c^2} + \frac{b}{a^2 c^2} + \frac{c}{a^2 b^2} - \frac{42}{14abc}$$

$$3 + \frac{a^3 + b^3 + c^3 - 3abc}{a^2 b^2 c^2}$$

$$\text{If } a + b + c = 0$$

$$\Rightarrow a^3 + b^3 + c^3 = 3abc$$

$$\Rightarrow 3$$

5. **Ans (C)**

$$I = \int_0^{10} [x] \cdot e^{[x]-x+1} dx$$

$$I = \int_0^1 0 dx + \int_1^2 1 \cdot e^{2-x} + \int_2^3 2 \cdot e^{3-x} + \dots + \int_9^{10} 9 \cdot e^{10-x} dx$$

$$\Rightarrow I = \sum_{n=0}^9 \int_n^{n+1} n \cdot e^{n+1-x} dx$$

$$= - \sum_{n=0}^9 n (e^{n+1-x})_n^{n+1}$$

$$= - \sum_{n=0}^9 n \cdot (e^0 - e^1)$$

$$= (e-1) \sum_{n=0}^9 n$$

$$= (e-1) \cdot \frac{9 \cdot 10}{2}$$

$$= 45(e-1)$$

6. **Ans (A)**

Given $y^2 = 4x$

Mirror image on $y = x \Rightarrow C : x^2 = 4y$

$$2x = 4 \cdot \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{x}{2}$$

$$\left. \frac{dy}{dx} \right|_{P(2,1)} = \frac{2}{2} = 1$$

Equation of tangent at (2, 1)

$$\Rightarrow y - 1 = 1(x - 2)$$

$$\Rightarrow x - y = 1$$

7. **Ans (B)**

$$\frac{dy}{dx} + (\tan x) y = \sin x; \quad 0 \leq x \leq \frac{\pi}{3}$$

$$I.F. = e^{\int \tan x dx} = e^{\ln \sec x} = \sec x$$

$$y \sec x = \int \tan x dx$$

$$y \sec x = \int \tan x dx$$

$$y \sec x = \ln |\sec x| + C$$

$$x = 0, y = 0 \Rightarrow \therefore C = 0$$

$$y \sec x = \ln |\sec x|$$

$$y = \cos x \cdot \ln |\sec x|$$

$$y|_{x=\frac{\pi}{4}} = \left(\frac{1}{\sqrt{2}} \right) \cdot \ln \sqrt{2}$$

$$y|_{x=\frac{\pi}{4}} = \frac{1}{2\sqrt{2}} \log_e 2$$

8. **Ans (D)**

$$A = \{2, 3, 4, 5, \dots, 30\}$$

$$(a, b) \simeq (c, d) \Rightarrow ad = bc$$

$$(4, 3) \simeq (c, d) \Rightarrow 4d = 3c$$

$$\Rightarrow \frac{4}{3} = \frac{c}{d}$$

$$\frac{c}{d} = \frac{4}{3} \text{ \& } c, d \in \{2, 3, \dots, 30\}$$

$$(c, d) = \{(4, 3), (8, 6), (12, 9), (16, 12), (20, 15), (24, 18), (28, 21)\}$$

$$\text{No. of ordered pair} = 7$$

9. **Ans (C)**

$$x^2 + y^2 + ax + 2ay + c = 0$$

$$2\sqrt{g^2 - c} = 2\sqrt{\frac{a^2}{4} - c} = 2\sqrt{2}$$

$$\Rightarrow \frac{a^2}{4} - c = 2 \quad \dots(1)$$

$$2\sqrt{f^2 - c} = 2\sqrt{a^2 - c} = 2\sqrt{5}$$

$$\Rightarrow a^2 - c = 5 \quad \dots(2)$$

(1) & (2)

$$\frac{3a^2}{4} = 3 \Rightarrow a = -2 \quad (a < 0)$$

$$\therefore c = -1$$

$$\text{Circle} \Rightarrow x^2 + y^2 - 2x - 4y - 1 = 0$$

$$\Rightarrow (x-1)^2 + (y-2)^2 = 6$$

$$\text{Given } x + 2y = 0 \Rightarrow m = -\frac{1}{2}$$

$$m_{\text{tangent}} = 2$$

Equation of tangent

$$\Rightarrow (y-2) = 2(x-1) \pm \sqrt{6}\sqrt{1+4}$$

$$\Rightarrow 2x - y \pm \sqrt{30} = 0$$

$$\text{Perpendicular distance from } (0, 0) = \left| \frac{\pm\sqrt{30}}{\sqrt{4+1}} \right| = \sqrt{6}$$

10. Ans (A)

$$\exp\left(\frac{(|z|+3)(|z|-1)}{|z|+1} \ln 2\right) \geq \log_{\sqrt{2}} |5\sqrt{7}+9i|$$

$$\Rightarrow 2^{\frac{(|z|+3)(|z|-1)}{|z|+1}} \geq \log_{\sqrt{2}}(16)$$

$$\Rightarrow 2^{\frac{(|z|+3)(|z|-1)}{|z|+1}} \geq 2^3$$

$$\Rightarrow \frac{(|z|+3)(|z|-1)}{|z|+1} \geq 3$$

$$\Rightarrow (|z|+3)(|z|-1) \geq 3(|z|+1)$$

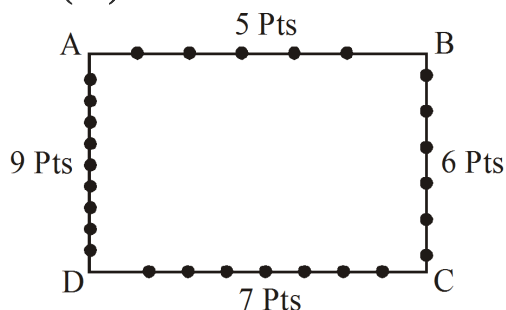
$$|z|^2 + 2|z| - 3 \geq 3|z| + 3$$

$$\Rightarrow |z|^2 + |z| - 6 \geq 0$$

$$\Rightarrow (|z|-3)(|z|+2) \geq 0 \Rightarrow |z|-3 \geq 0$$

$$\Rightarrow |z| \geq 3 \Rightarrow |z|_{\min} = 3$$

11. Ans (D)



$$\alpha = \text{Number of triangles}$$

$$\alpha = 5 \cdot 6 \cdot 7 + 5 \cdot 7 \cdot 9 + 5 \cdot 6 \cdot 9 + 6 \cdot 7 \cdot 9$$

$$= 210 + 315 + 270 + 378$$

$$= 1173$$

$$\beta = \text{Number of Quadrilateral}$$

$$\beta = 5 \cdot 6 \cdot 7 \cdot 9 = 1890$$

$$\beta - \alpha = 1890 - 1173 = 717$$

12. Ans (A)

$$y^2 = 3x^2$$

and $x^2 + y^2 = 4b$

Solve both we get

so $x^2 = b$

$$\frac{x^2}{16} + \frac{3x^2}{b^2} = 1$$

$$\frac{b}{16} + \frac{3}{b} = 1$$

$$b^2 - 16b + 48 = 0$$

$$(b-12)(b-4) = 0$$

$$b = 12, b > 4$$

13. Ans (C)

$$\sin^{-1} \frac{3x}{5} + \sin^{-1} \frac{4x}{5} = \sin^{-1} x$$

$$\sin^{-1} \left(\frac{3x}{5} \sqrt{1 - \frac{16x^2}{25}} + \frac{4x}{5} \sqrt{1 - \frac{9x^2}{25}} \right) = \sin^{-1} x$$

$$\frac{3x}{5} \sqrt{1 - \frac{16x^2}{25}} + \frac{4x}{5} \sqrt{1 - \frac{9x^2}{25}} = x$$

$$x = 0, 3\sqrt{25 - 16x^2} + 4\sqrt{25 - 9x^2} = 25$$

$$4\sqrt{25 - 9x^2} = 25 - 3\sqrt{25 - 16x^2} \quad \text{squaring}$$

we get

$$16(25 - 9x^2) = 625 + 9(25 - 16x^2) - 150\sqrt{25 - 16x^2}$$

$$400 = 625 + 225 - 150\sqrt{25 - 16x^2}$$

$$\sqrt{25 - 16x^2} = 3 \Rightarrow 25 - 16x^2 = 9$$

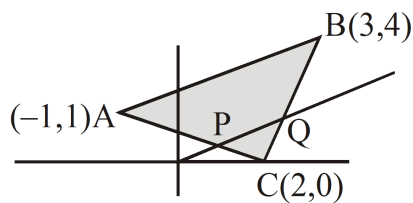
$$\Rightarrow x^2 = 1$$

Put $x = 0, 1, -1$ in the original equation

We see that all values satisfy the original equation.

Number of solution = 3

14. Ans (B)



$$P \equiv (x_1, mx_1)$$

$$Q \equiv (x_2, mx_2)$$

$$A_1 = \frac{1}{2} \begin{vmatrix} 3 & 4 & 1 \\ 2 & 0 & 1 \\ -1 & 1 & 1 \end{vmatrix} = \frac{13}{2}$$

$$A_2 = \frac{1}{2} \begin{vmatrix} x_1 & mx_1 & 1 \\ x_2 & mx_2 & 1 \\ 2 & 0 & 1 \end{vmatrix}$$

$$A_2 = \frac{1}{2} |2(mx_1 - mx_2)| = m|x_1 - x_2|$$

$$A_1 = 3A_2 \Rightarrow \frac{13}{2} = 3m|x_1 - x_2|$$

$$\Rightarrow |x_1 - x_2| = \frac{16}{6m}$$

$$AC : x + 3y = 2$$

$$BC : y = 4x - 8$$

$$P : x + 3y = 2 \text{ \& } y = mx \Rightarrow x_1 = \frac{2}{1+3m}$$

$$Q : y = 4x - 8 \text{ \& } y = mx \Rightarrow x_2 = \frac{8}{4-m}$$

$$|x_1 - x_2| = \left| \frac{2}{1+3m} - \frac{8}{4-m} \right|$$

$$= \left| \frac{-26m}{(1+3m)(4-m)} \right| = \frac{26m}{(3m+1)|m-4|}$$

$$= \frac{26m}{(3m+1)(4-m)}$$

$$|x_1 - x_2| = \frac{13}{6m}$$

$$\frac{26m}{(3m+1)(4-m)} = \frac{13}{6m}$$

$$\Rightarrow 12m^2 = -(3m+1)(m-4)$$

$$\Rightarrow 12m^2 = -(3m^2 - 11m - 4)$$

$$\Rightarrow 15m^2 - 11m - 4 = 0$$

$$\Rightarrow 15m^2 - 15m + 4m - 4 = 0$$

$$\Rightarrow (15m+4)(m-1) = 0$$

$$\Rightarrow m = 1$$

15. Ans (A)

$$f(x) = 3\ln(x-1) - 3\ln(x+1) - \frac{2}{x-1}$$

$$f'(x) = \frac{3}{x-1} - \frac{3}{x+1} + \frac{2}{(x-1)^2}$$

$$f'(x) = \frac{4(2x-1)}{(x-1)^2(x+1)}$$

$$f'(x) \geq 0$$

$$\Rightarrow x \in (-\infty, -1) \cup \left[\frac{1}{2}, 1\right) \cup (1, \infty)$$

16. Ans (A)

$$\ln f(x+1) = \ln(xf(x))$$

$$\ln f(x+1) = \ln x + \ln f(x)$$

$$\Rightarrow g(x+1) = \ln x + g(x)$$

$$\Rightarrow g(x+1) - g(x) = \ln x$$

$$\Rightarrow g''(x+1) - g''(x) = -\frac{1}{x^2}$$

$$\text{Put } x = 1, 2, 3, 4$$

$$g''(2) - g''(1) = -\frac{1}{1^2} \quad \dots(1)$$

$$g''(3) - g''(2) = -\frac{1}{2^2} \quad \dots(2)$$

$$g''(4) - g''(3) = -\frac{1}{3^2} \quad \dots(3)$$

$$g''(5) - g''(4) = -\frac{1}{4^2} \quad \dots(4)$$

$$\text{Add all the equation we get}$$

$$g''(5) - g''(1) = -\frac{1}{1^2} - \frac{1}{2^2} - \frac{1}{3^2} - \frac{1}{4^2}$$

$$|g''(5) - g''(1)| = \frac{205}{144}$$

17. Ans (C)

$$\int_0^1 (x^2 + bx + c) dx = 1$$

$$\frac{1}{3} + \frac{b}{2} + c = 1 \Rightarrow \frac{b}{2} + c = \frac{2}{3}$$

$$3b + 6c = 4 \quad \dots(1)$$

$$P(2) = 5$$

$$4 + 2b + c = 5$$

$$2b + c = 1 \quad \dots(2)$$

$$\text{From (1) \& (2)}$$

$$b = \frac{2}{9} \text{ \& } c = \frac{5}{9}$$

$$9(b+c) = 7$$

18. Ans (B)

(3,5,7) satisfy the line L_1

$$\frac{3-a}{\ell} = \frac{5-2}{3} = \frac{7-b}{4}$$

$$\frac{3-a}{\ell} = 1 \quad \& \quad \frac{7-b}{4} = 1$$

$$a + \ell = 3 \quad \dots(1) \quad \& \quad b = 3 \quad \dots(2)$$

$$\vec{v}_1 = \langle 4, 3, 8 \rangle - \langle 3, 5, 7 \rangle$$

$$\vec{v}_1 = \langle 1, -2, 1 \rangle$$

$$\vec{v}_2 = \langle \ell, 3, 4 \rangle$$

$$\vec{v}_1 \cdot \vec{v}_2 = 0 \Rightarrow \ell - 6 + 4 = 0 \Rightarrow \ell = 2$$

$$a + \ell = 3 \Rightarrow a = 1$$

$$L_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$$

$$L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$$

$$A = \langle 1, 2, 3 \rangle$$

$$B = \langle 2, 4, 5 \rangle$$

$$\vec{AB} = \langle 1, 2, 2 \rangle$$

$$\vec{p} = 2\hat{i} + 3\hat{j} + 4\hat{k}$$

$$\vec{q} = 3\hat{i} + 4\hat{j} + 5\hat{k}$$

$$\vec{p} \times \vec{q} = -\hat{i} + 2\hat{j} - \hat{k}$$

$$\text{Shortest distance} = \frac{|\vec{AB} \cdot (\vec{p} \times \vec{q})|}{|\vec{p} \times \vec{q}|} = \frac{1}{\sqrt{6}}$$

19. Ans (B)

$$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}, \quad x \in (0, \infty)$$

put $y = vx$

$$x \frac{dv}{dx} + v = \frac{v^2 - 1}{2v}$$

$$\frac{2v}{v^2 + 1} dv = -\frac{dx}{x}$$

Integrate,

$$\ell n(v^2 + 1) = -\ell nx + C$$

$$\ell n \left(\frac{y^2}{x^2} + 1 \right) = -\ell nx + C$$

$$\text{put } x = 1, y = 1, C = \ell n 2$$

$$\ell n \left(\frac{y^2}{x^2} + 1 \right) = -\ell nx + \ell n 2$$

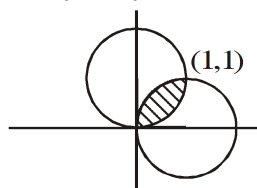
$$\Rightarrow x^2 + y^2 - 2x = 0 \quad (\text{Curve } C_1)$$

Similarly,

$$\frac{dy}{dx} = \frac{2xy}{x^2 - y^2}$$

Put $y = vx$

$$x^2 + y^2 - 2y = 0$$



$$\text{required area} = 2 \int_0^1 (\sqrt{2x - x^2} - x) dx = \frac{\pi}{2} - 1$$

20. Ans (B)

$$\vec{r} \times \vec{a} = \vec{b} \times \vec{r} \Rightarrow \vec{r} \times (\vec{a} + \vec{b}) = 0$$

$$\vec{r} = \lambda(\vec{a} + \vec{b}) \Rightarrow \vec{r} = \lambda(\hat{i} + 2\hat{j} - 3\hat{k} + 2\hat{i} - 3\hat{j} + 5\hat{k})$$

$$\vec{r} = \lambda(3\hat{i} - \hat{j} + 2\hat{k}) \quad \dots(1)$$

$$\vec{r} \cdot (\alpha\hat{i} + 2\hat{j} + \hat{k}) = 3$$

$$\text{Put } \vec{r} \text{ from (1) } \alpha\lambda = 1 \quad \dots(2)$$

$$\vec{r} \cdot (2\hat{i} + 5\hat{j} - \alpha\hat{k}) = -1$$

$$\text{Put } \vec{r} \text{ from (1) } 2\lambda\alpha - \lambda = 1 \quad \dots(3)$$

Solve (2) & (3)

$$\alpha = 1, \lambda = 1$$

$$\Rightarrow \vec{r} = 3\hat{i} - \hat{j} + 2\hat{k}$$

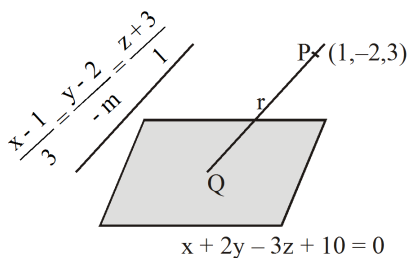
$$|\vec{r}|^2 = 14 \quad \& \quad \alpha = 1$$

$$\alpha + |\vec{r}|^2 = 15$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (2)



DC of line

$$\equiv \left(\frac{3}{\sqrt{m^2 + 10}}, \frac{-m}{\sqrt{m^2 + 10}}, \frac{1}{\sqrt{m^2 + 10}} \right)$$

$$Q \equiv \left(1 + \frac{3r}{\sqrt{m^2 + 10}}, -2 + \frac{-mr}{\sqrt{m^2 + 10}}, 3 + \frac{r}{\sqrt{m^2 + 10}} \right)$$

Q lies on $x + 2y - 3z + 10 = 0$

$$1 + \frac{3r}{\sqrt{m^2 + 10}} - 4 - \frac{2mr}{\sqrt{m^2 + 10}} - 9 - \frac{3r}{\sqrt{m^2 + 10}} + 10 = 0$$

$$\Rightarrow \frac{r}{\sqrt{m^2 + 10}} (3 - 2m - 3) = 2$$

$$\Rightarrow \frac{r}{\sqrt{m^2 + 10}} (-2m) = 2$$

$$r^2 m^2 = m^2 + 10$$

$$\frac{7}{2} m^2 = m^2 + 10 \Rightarrow \frac{5}{2} m^2 = 10 \Rightarrow m^2 = 4$$

$$|m| = 2$$

2. Ans (5)

$$\sigma^2 = \frac{n_1 \sigma_1^2 + n_2 \sigma_2^2}{n_1 + n_2} + \frac{n_1 n_2}{(n_1 + n_2)^2} (\bar{x}_1 - \bar{x}_2)^2$$

$$n_1 = 10, n_2 = n, \sigma_1^2 = 2, \sigma_2^2 = 1$$

$$\bar{x}_1 = 2, \bar{x}_2 = 3, \sigma^2 = \frac{17}{9}$$

$$\frac{17}{9} = \frac{10 \times 2 + n}{n + 10} + \frac{10n}{(n + 10)^2} (3 - 2)^2$$

$$\Rightarrow \frac{17}{9} = \frac{(n + 20)(n + 10) + 10n}{(n + 10)^2}$$

$$\Rightarrow 17n^2 + 1700 + 340n = 90n + 9(n^2 + 30n + 200)$$

$$\Rightarrow 8n^2 - 20n - 100 = 0$$

$$2n^2 - 5n - 25 = 0$$

$$\Rightarrow (2n + 5)(n - 5) = 0 \Rightarrow n = \frac{-5}{2}, 5$$

↓

(Rejected)

Hence $n = 5$

3. Ans (1)

$$A = XB$$

$$\begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{3} a_1 \\ \sqrt{3} a_2 \end{bmatrix} = \begin{bmatrix} b_1 - b_2 \\ b_1 + k b_2 \end{bmatrix}$$

$$b_1 - b_2 = \sqrt{3} a_1 \quad \dots(1)$$

$$b_1 + k b_2 = \sqrt{3} a_2 \quad \dots(2)$$

$$\text{Given, } a_1^2 + a_2^2 = \frac{2}{3} (b_1^2 + b_2^2)$$

$$(1)^2 + (2)^2$$

$$(b_1 + b_2)^2 + (b_1 + k b_2)^2 = 3(a_1^2 + a_2^2)$$

$$a_1^2 + a_2^2 = \frac{2}{3} b_1^2 + \frac{(1 + k^2)}{3} b_2^2 + \frac{2}{3} b_1 b_2 (k - 1)$$

$$\text{Given, } a_1^2 + a_2^2 = \frac{2}{3} b_1^2 + \frac{2}{3} b_2^2$$

On comparing we get

$$\frac{k^2 + 1}{3} = \frac{2}{3} \Rightarrow k^2 + 1 = 2$$

$$\Rightarrow k = \pm 1 \quad \dots(3)$$

$$\& \frac{2}{3} (k - 1) = 0 \Rightarrow k = 1 \quad \dots(4)$$

From both we get $k = 1$

4. Ans (6)

$$\int \frac{(x^2-1)dx}{(x^4+3x^2+1)\tan^{-1}\left(x+\frac{1}{x}\right)} + \int \frac{dx}{x^4+3x^2+1}$$

$$\int \frac{\left(1-\frac{1}{x^2}\right)dx}{\left(\left(x+\frac{1}{x}\right)^2+1\right)\tan^{-1}\left(x+\frac{1}{x}\right)} + \frac{1}{2} \int \frac{(x^2+1)-(x^2-1)dx}{x^4+3x^2+1}$$

Put $\tan^{-1}\left(x+\frac{1}{x}\right) = t$

$$\int \frac{dt}{t} + \frac{1}{2} \int \frac{\left(1+\frac{1}{x^2}\right)dx}{\left(x-\frac{1}{x}\right)^2+5} - \frac{1}{2} \int \frac{\left(1-\frac{1}{x^2}\right)dx}{\left(x+\frac{1}{x}\right)^2+1}$$

Put $x - \frac{1}{x} = y, x + \frac{1}{x} = z$

$$\log_e t + \frac{1}{2} \int \frac{dy}{y^2+5} - \frac{1}{2} \int \frac{dz}{z^2+1}$$

$$= \log_e \tan^{-1}\left(x+\frac{1}{x}\right) + \frac{1}{2\sqrt{5}} \tan^{-1}\left(\frac{x^2-1}{\sqrt{5}x}\right)$$

$$- \frac{1}{2} \tan^{-1}\left(\frac{x^2+1}{x}\right) + C$$

$$\alpha = 1, \beta = \frac{1}{2\sqrt{5}}, \gamma = \frac{1}{\sqrt{5}}, \delta = \frac{-1}{2}$$

or

$$\alpha = 1, \beta = \frac{-1}{2\sqrt{5}}, \gamma = \frac{-1}{\sqrt{5}}, \delta = \frac{-1}{2}$$

$$10(\alpha + \beta\gamma + \delta) = 10\left(1 + \frac{1}{10} - \frac{1}{2}\right) = 6$$

5. Ans (1)

$$g[f(x)] = \begin{cases} f(x) + 1 & f(x) < 0 \\ (f(x) - 1)^2 + b & f(x) \geq 0 \end{cases}$$

$$g[f(x)] = \begin{cases} x + a + 1 & x + a < 0 \& x < 0 \\ |x - 1| + 1 & |x - 1| < 0 \& x \geq 0 \\ (x + a - 1)^2 + b & x + a \geq 0 \& x < 0 \\ (|x - 1| - 1)^2 + b & |x - 1| \geq 0 \& x \geq 0 \end{cases}$$

$$g[f(x)] = \begin{cases} x + a + 1 & x \in (-\infty, -a) \& x \in (-\infty, 0) \\ |x - 1| + 1 & x \in \phi \\ (x + a - 1)^2 + b & x \in [-a, \infty) \& x \in (-\infty, 0) \\ (|x - 1| - 1)^2 + b & x \in \mathbb{R} \& x \in [0, \infty) \end{cases}$$

$$g[f(x)] = \begin{cases} x + a + 1 & x \in (-\infty, -a) \\ (x + a - 1)^2 + b & x \in [-a, 0) \\ (|x - 1| - 1)^2 + b & x \in [0, \infty) \end{cases}$$

$g(f(x))$ is continuous

at $x = -a$ & at $x = 0$

$$1 = b + 1 \quad \& \quad (a - 1)^2 + b = b$$

$$b = 0 \quad \& \quad a = 1$$

$$\Rightarrow a + b = 1$$

6. Ans (14)

$$a^2 = \frac{b}{16} \Rightarrow \frac{1}{b} = \frac{1}{16a^2}$$

$$\frac{2}{b} = \frac{1}{a} + 6$$

$$\frac{1}{8a^2} = \frac{1}{a} + 6$$

$$\frac{1}{a^2} - \frac{8}{a} - 48 = 0$$

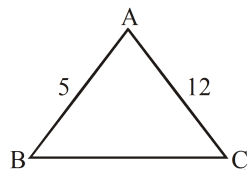
$$\frac{1}{a} = 12, -4 \Rightarrow a = \frac{1}{12}, -\frac{1}{4}$$

$$a = \frac{1}{12}, a > 0$$

$$b = 16a^2 = \frac{1}{9}$$

$$\Rightarrow 72(a + b) = 6 + 8 = 14$$

7. Ans (15)



$$\Delta = \frac{1}{2} \cdot 5 \cdot 12 \cdot \sin A = 30$$

$$\sin A = 1$$

$$A = 90^\circ \Rightarrow BC = 13$$

$$BC = 2R = 13$$

$$r = \frac{\Delta}{S} = \frac{30}{15} = 2$$

$$2R + r = 15$$

8. Ans (6)

$$A =$$

$$\sum_{k=0}^n {}^nC_k \left[\left(-\frac{1}{2}\right)^k + \left(-\frac{3}{4}\right)^k + \left(-\frac{7}{8}\right)^k + \left(-\frac{15}{16}\right)^k + \left(-\frac{31}{32}\right)^k \right]$$

$$A =$$

$$\left(1 - \frac{1}{2}\right)^n + \left(1 - \frac{3}{4}\right)^n + \left(1 - \frac{7}{8}\right)^n + \left(1 - \frac{15}{16}\right)^n + \left(1 - \frac{31}{32}\right)^n$$

$$A = \frac{1}{2^n} + \frac{1}{4^n} + \frac{1}{8^n} + \frac{1}{16^n} + \frac{1}{32^n}$$

$$A = \frac{1}{2^n} \left(\frac{1 - \left(\frac{1}{2^n}\right)^5}{1 - \frac{1}{2^n}} \right) \Rightarrow A = \frac{\left(1 - \frac{1}{2^{5n}}\right)}{(2^n - 1)}$$

$$(2n - 1)A = 1 - \frac{1}{2^{5n}}, \text{ Given } 63A = 1 - \frac{1}{2^{30}}$$

$$\text{Clearly } 5n = 30$$

$$n = 6$$

9. Ans (28)

$$\vec{c} = \lambda(\vec{a} \times \vec{b})$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix}$$

$$(\vec{a} \times \vec{b}) = 3\hat{i} - 2\hat{j} + \hat{k}$$

$$\vec{c} \cdot (\hat{i} + \hat{j} + 3\hat{k}) = \lambda(3\hat{i} - 2\hat{j} + \hat{k}) \cdot (\hat{i} + \hat{j} + 3\hat{k})$$

$$\Rightarrow \lambda(4) = 8 \Rightarrow \lambda = 2$$

$$\vec{c} = 2(\vec{a} \times \vec{b})$$

$$\vec{c} \cdot (\vec{a} \times \vec{b}) = 2|\vec{a} \times \vec{b}|^2 = 28$$

10. Ans (16)

$$S_n(x) = (2+3+6+11+18+27+\dots+n\text{-terms})\log_a x$$

$$\text{Let } S_1 = 2 + 3 + 6 + 11 + 18 + 27 + \dots + T_n$$

$$S_1 = 2 + 3 + 6 + \dots + T_n$$

$$\dots$$

$$T_n = 2 + 1 + 3 + 5 + \dots + n \text{ terms}$$

$$T_n = 2 + (n - 1)^2$$

$$S_1 = \Sigma T_n = 2n + \frac{(n - 1)n(2n - 1)}{6}$$

$$\Rightarrow S_n(x) = \left(2n + \frac{n(n - 1)(2n - 1)}{6}\right) \log_a x$$

$$S_{24}(x) = 1093 \text{ (Given)}$$

$$\log_a x \left(48 + \frac{23 \cdot 24 \cdot 47}{6}\right) = 1093$$

$$\log_a x = \frac{1}{4} \dots (1)$$

$$S_{12}(2x) = 265$$

$$S_{12}(2x) = 265$$

$$\log_a(2x) \left(24 + \frac{11 \cdot 12 \cdot 23}{6}\right) = 265$$

$$\log_a 2x = \frac{1}{2} \dots (2)$$

$$(2) - (1)$$

$$\log_a 2x - \log_a x = \frac{1}{4}$$

$$\log_a 2 = \frac{1}{4} \Rightarrow a = 16$$